

Premalignant Conditions

- **Leukoplakia:** White patches in the mouth that may become cancerous.
- **Erythroplakia:** Red patches in the mouth that have a higher chance of becoming cancerous.

Six Ss (Risk factors for cancer)

- **Smoking:** Inhaling tobacco smoke.
- **Syphilis:** A sexually transmitted bacterial infection.
- **Spices:** Use of certain spicy foods that may irritate oral mucosa.
- **Spirits:** Alcohol consumption.
- **Sharp tooth:** A tooth that irritates the mouth lining, increasing cancer risk.
- **Sepsis (betel nut/tobacco):** Infection due to chewing betel nut or tobacco.

Types of Carcinoma

- **Non-healing ulcer:** An ulcer that doesn't heal over time.
- **Proliferative growth:** Abnormal growth of tissue.
- **Frozen tongue:** A tongue that becomes stiff, usually due to cancer.
- **Fissure variety:** Cancer forming cracks or grooves in the tongue or mouth.

Histological Types

- **Squamous cell carcinoma:** Cancer originating from squamous epithelial cells (most common).
- **Adenocarcinoma:** Cancer originating from glandular tissue, such as minor salivary glands.
- **Melanomas:** Skin cancer that can also affect mucosal surfaces.

Clinical Features

- **Ankyloglossia:** A condition where the tongue is restricted in movement due to a short frenulum.
- **Foetororis (halitosis):** Foul-smelling breath.

Spread

- **Lymphatic:** Spread of cancer to lymph nodes.
- **Blood:** Spread through the bloodstream, usually in late stages.

Investigations

- **Edge biopsy:** A tissue sample taken from the edge of a lesion.

- **FNAC of lymph nodes:** Fine needle aspiration cytology, a test to check for cancer in lymph nodes.
- **Indirect laryngoscopy:** A procedure to view the throat and larynx.
- **CT scan:** A detailed imaging scan of the area.
- **Chest X-ray:** An imaging test used to look for lung involvement.

Treatment

- **Glossectomy:** Surgical removal of part or all of the tongue.
- **Commando's operation:** A type of surgery that involves removing the tongue and surrounding tissues.
- **Teletherapy:** A type of external radiation therapy.
- **Interstitial:** Radiation therapy delivered directly to the tumor site.

Secondary Lymph Nodes

- **Prophylactic block dissection:** Preventive surgery to remove lymph nodes at risk of cancer spread.
- **En bloc radical neck dissection:** Extensive surgery to remove lymph nodes and surrounding tissues in the neck.
- **Palliative radiotherapy/chemotherapy:** Treatment aimed at relieving symptoms rather than curing cancer.

Prognosis

- **Cachexia:** A condition of extreme weight loss and muscle wasting, often associated with cancer.
- **Aspiration:** Inhalation of food or liquid into the lungs, leading to choking or pneumonia.
- **Hemorrhage:** Excessive bleeding.
- **Asphyxia:** Lack of oxygen leading to suffocation.

Aphthous Ulcers

Recurrent: Occurs repeatedly over time.

Erythematous: Red or inflamed.

Chlorhexidine gluconate: Antiseptic mouthwash used to kill bacteria.

Choline salicylate: Gel used for pain relief.

Placebo effect: Improvement due to belief in treatment rather than the treatment itself.

Dental Ulcer (Traumatic)

Indurated: Hardened or thickened.

Ill-fitting dentures: Dentures that do not fit properly, causing irritation.

Chronic Non-Specific Ulcer

Indurated: Hardened or thickened.

Cauterization: The process of burning tissue to remove or close a wound.

Syphilitic or Gummatous Ulcer

Snail track ulcers: Ulcers with an elongated, irregular, and deep appearance.

Gumma: A soft, tumor-like growth associated with tertiary syphilis.

Chancre: A painless ulcer found in the primary stage of syphilis.

Mucous patches: Lesions occurring in syphilis, characterized by white, velvety plaques.

Punched-out edges: Characteristic appearance of syphilitic ulcers.

Wash leather slough: A dry, leathery slough or shedding of dead tissue.

Tubercular Ulcer of Tongue

Fulminating: Occurring suddenly with severe symptoms.

Serous discharge: Fluid that is clear or yellowish, typically associated with inflammation.

Matted lymph nodes: Lymph nodes that are swollen and stuck together.

Systemic Diseases

Pemphigus: An autoimmune disorder causing blisters and sores on the skin and mucous membranes.

Systemic lupus erythematosus (SLE): An autoimmune disease that causes inflammation and damage to various body systems.

Lichen planus: An inflammatory condition that affects the skin and mucous membranes.

Herpetic Ulcer

Neuralgic pain: Pain associated with nerve damage or irritation.

Vesicles: Small fluid-filled blisters.

Herpes zoster: A viral infection caused by the varicella-zoster virus, leading to painful skin rashes.

Glossitis Causing Ulcers

Hyperemic: Excessively red due to increased blood flow.

Smoker's ulcer: An ulcer caused by chronic irritation from smoking.

Post-Pertussis Ulcer (Whooping Cough)

Frenulum: The small fold of tissue that connects the underside of the tongue to the floor of the mouth.

Carcinoma of the Cheek

Aetiology

- **Hyperplastic candidiasis:** Thickened fungal infection causing white patches.
- **Submucosal fibrosis:** Chronic disease with stiffening of oral tissues.
- **Betel nut chewing:** Habit of chewing betel nut (areca) with or without tobacco.

Pathological Types

- **Exophytic growth:** Outward-growing lesion like verrucous carcinoma.
- **Infiltrative lesion:** Tumor invading surrounding tissues.

Clinical Features

- **Everted edges:** Outward-turned edges of the ulcer.
- **Induration:** Hardening of tissues.
- **Retromolar trigone:** Area behind the last molar.
- **Trismus:** Limited mouth opening due to muscle involvement.
- **Pterygoid:** Muscles involved in jaw movement.
- **Halitosis:** Foul-smelling breath.

Investigations

- **Wedge biopsy:** Sampling the lesion edge for diagnosis.
- **Keratin pearls:** Histological finding typical of squamous cell carcinoma.
- **OPG:** Orthopantomogram, an X-ray of the mandible.
- **FNAC:** Fine needle aspiration cytology to examine lymph nodes.
- **Secondaries:** Distant metastases of cancer.

Treatment

Surgery

- **Split skin graft (SSG):** Thin layer of skin transplanted to cover defects.
- **PMMC flap:** Pectoralis major myocutaneous flap, a muscle and skin graft for reconstruction.

Radiotherapy

- **External radiotherapy:** Radiation applied from outside the body.
- **Interstitial radiotherapy:** Radiation delivered inside the tissue using radioactive wires.
- **Caesium 137/Iridium 192:** Radioactive isotopes used in treatment.
- **Commissure:** Corner of the mouth.

Chemotherapy

- **Intra-arterial:** Drug delivery directly into an artery.

Advanced Carcinoma of the Cheek

- **Deltpectoral flap:** Tissue graft from the chest region near the shoulder.
- **Forehead flap:** Skin graft based on the superficial temporal artery.
- **Mandible reconstruction:** Rebuilding the jaw using bone grafts or synthetic materials.
- **Free flaps (microvascular):** Tissue with its blood supply transferred to another site.

Lingual Thyroid

Description

- **Foramen caecum:** Depression at the back of the tongue marking thyroid's embryological origin.

Development

- **Thyroglossal bud:** Embryonic structure that forms the thyroid gland.
- **Arrested descent:** Stopping of the thyroid's migration, causing ectopic tissue.

Important Note

- **Reimplantation:** Placing removed thyroid tissue elsewhere to maintain hormone production.
- **Thyroid hormone deficiency:** Lack of thyroid hormones, leading to hypothyroidism.

Potential Complications

- **Haemorrhage:** Excessive bleeding.
- **Respiratory obstruction:** Blockage causing breathing difficulty.

- **Dysphagia:** Difficulty swallowing.
- **Speech impairment:** Trouble with speaking.

Diagnosis

1. **Radioisotope study:** Imaging technique to check iodine absorption and thyroid status.
2. **Ultrasound:** Imaging to locate thyroid tissue and confirm absence in the neck.

Treatment

1. **L-thyroxine:** Synthetic thyroid hormone to shrink the lingual thyroid.
2. **Surgical excision:** Removal of the lingual thyroid when necessary.
3. **Radioisotope therapy:** Destruction of thyroid tissue using radioactive iodine.

Key Points

- **Ectopic thyroid:** Abnormal thyroid tissue not in its usual location.
- **Careful management:** Essential to avoid lifelong thyroid hormone deficiency.

Deep Vein Thrombosis (DVT)

Definition

- **Thrombus:** Blood clot.
- **Deep venous system:** Network of veins located deep within the body.

Causes (Etiology)

- **Venous stasis:** Slow blood flow, often from immobility.
- **Hypercoagulability:** Increased tendency for clotting (e.g., in pregnancy, cancer).
- **Endothelial injury:** Damage to the blood vessel lining.

Risk Factors

- **Prolonged immobility:** Long travel or hospitalization.
- **Obesity:** Excess weight increases pressure on veins.
- **Oral contraceptives:** Hormonal medications raising clot risk.
- **Smoking:** Contributes to vascular damage.

Pathophysiology

- **Stasis:** Blood pooling in veins.
- **Hypercoagulability:** Increased clot formation tendency.
- **Obstruction:** Clots block blood flow.

Clinical Features

- **Homan's Sign:** Pain on dorsiflexion of the foot.
- **Redness and warmth:** Local inflammation signs.

Complications

1. **Pulmonary embolism (PE):** Clot traveling to the lungs, causing a blockage.
2. **Post-thrombotic syndrome:** Long-term swelling and pain in the affected limb.
3. **Recurrent DVT:** Clots reappearing after initial treatment.

Investigations

- **D-dimer test:** Blood test for clot breakdown products.
- **Doppler ultrasound:** Imaging to locate blood clots.
- **CT/MRI venography:** Advanced imaging techniques.

Treatment

1. General Measures

- **Elastic stockings:** Specialized socks to improve blood flow.
- **Leg elevation:** Raising legs to decrease swelling.
- **Physiotherapy:** Exercises to restore mobility.

2. Medications

- **Heparin:** Anticoagulant for immediate action.
- **Warfarin:** Long-term oral blood thinner.
- **Thrombolytics:** Clot-dissolving drugs in severe cases.

3. Surgical Management

- **Thrombectomy:** Surgical removal of a clot.
- **IVC filter:** Device to catch clots before they reach the lungs.

Prevention

- **Early mobilization:** Moving soon after surgery or immobility.
- **Compression stockings:** Prevention during travel or prolonged sitting.
- **Hydration:** Adequate water intake.

Key Points

- **INR:** Blood test to monitor warfarin therapy (goal: 2-3).
- **Prompt diagnosis:** Essential to avoid severe complications like PE.

Secondaries in the Neck with Occult Primary

Definition

- Metastatic neck lymph nodes where the primary tumor's origin is unknown.

Causes

1. **Head and Neck Cancers**
 - Nasopharyngeal carcinoma.
 - Oropharyngeal carcinoma (e.g., tonsil, tongue base).
 - Hypopharyngeal carcinoma.
 - Laryngeal carcinoma.
2. **Upper Aerodigestive Tract Cancers**
 - Esophageal carcinoma.
 - Tracheal carcinoma.
3. **Thyroid Cancer**
 - Papillary carcinoma.
 - Medullary carcinoma.
4. **Salivary Gland Cancer**
 - Originating from minor salivary glands.
5. **Distant Primary Tumors**
 - Lung carcinoma.
 - Breast carcinoma.
 - Gastrointestinal tract cancers (e.g., stomach).
6. **Skin Cancer**
 - Melanoma.
 - Squamous cell carcinoma.
7. **Lymphoma**
 - Hodgkin's lymphoma.
 - Non-Hodgkin's lymphoma.

Management

1. Detailed History and Examination

- **History**
 - Smoking, alcohol use, and family cancer history.
 - Symptoms like dysphagia, hoarseness, or weight loss.
- **Examination**
 - Assess neck nodes for size, location, and consistency.
 - Comprehensive examination of oral cavity, nasopharynx, and oropharynx.

2. Investigations

- **Basic Tests**
 - **FNAC:** Confirms metastasis and cell type.
 - **Biopsy:** Excisional or core biopsy for inconclusive FNAC.
- **Imaging**
 - **CT/MRI:** Evaluates primary site and node involvement.
 - **PET-CT:** Detects hidden primary tumors.
- **Endoscopy**
 - **Panendoscopy:** Visualizes oropharynx, nasopharynx, hypopharynx